

AI-POWERED SUPPLY CHAIN & PROCUREMENT ANALYTICS

Business Requirements Document

Project Name	AI-Powered Supply Chain & Procurement Analytics
Document Name	Business Requirements Document (BRD)
Version	1.0
Date	
Prepared By	Business Analyst (Portfolio Project Author)
Document Status	Draft for Stakeholder Review

Primary Data Source: SupplyChain_Analytics_10000.csv · ~10,000 order records · 2024–2025

BRD = What the business needs and why. A separate FRD will later define how the solution must function.

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1. Document Information

Field	Value
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This document is a Business Analyst portfolio deliverable. All stakeholder references throughout this document are organizational roles, not named individuals.

2. Executive Summary

Business situation: The organization captures detailed supply-chain transaction data — suppliers, products, warehouses, customers, regions, transportation modes, orders, inventory, cost, and profitability — but has no centralized analytical capability to turn that data into decision-support.

Business problem: Supply-chain and procurement stakeholders lack integrated visibility into performance, cannot easily identify operational issues as they emerge, and have limited ability to act proactively on risk. Decisions are made from fragmented, manually assembled information rather than a consistent evidence base.

Proposed business solution: This project will deliver an integrated business analysis and analytics capability covering supplier performance, order fulfillment, delivery and logistics, inventory, cost and profitability, standardized KPIs, management reporting, and predictive insight into delivery risk and demand — all built on the organization's existing historical transaction data (approximately 10,000 order records, 2024–2025).

Expected value: Stakeholders gain a consistent, evidence-based view of supply-chain performance, an earlier line of sight into operational risk, and analytical findings translated into actionable recommendations. Predictive components are historical-pattern-based decision support, not guarantees of future outcomes.

3. Business Context

The organization operates a global supply chain connecting 10 suppliers, 10 warehouses, 25 products across 5 categories, and customers across 27 countries in 5 regions, using 4 transportation modes (Road, Air, Sea, Rail). Suppliers, logistics, inventory, order fulfillment, and cost are interdependent: a delay at one supplier can affect warehouse inventory positions and, ultimately, customer delivery outcomes; cost and profitability are shaped by the combined effect of sourcing, transportation, and fulfillment decisions.

Order-level transaction records exist for January 2024 through December 2025, capturing order, shipment and delivery dates, cost, quantity, delivery time, inventory level, order status, and profit. Of the approximately 10,000 orders on record, 82.7% are Delivered, 8.2% are In Transit, 5.9% are Delayed, and 3.2% are Cancelled.

This data is captured consistently but is not currently organized into a form that supports comparative supplier evaluation, delivery-risk monitoring, cost analysis, or forward-looking forecasting. The current analytical challenge is not a lack of data, but a lack of a structured, repeatable way to use it — creating a clear need for data-driven, rather than ad hoc, decision-making.

4. Business Problem

Supply-chain and procurement stakeholders lack integrated visibility into supplier, delivery, inventory, cost, and profitability performance. This makes it difficult to identify operational issues as they emerge, limits proactive risk identification, and results in fragmented, inconsistent decision-making across functions.

This is a business visibility and decision-support problem, not a request for a specific tool. The underlying data already exists in the organization's records; what is missing is a structured, repeatable way to analyze and act on it.

5. Business Impact

The current gap in integrated visibility has the following potential impact across the business. No financial figures are estimated here beyond what the dataset itself shows (e.g., current order-status proportions); this section describes the nature of the impact, not a quantified cost of inaction.

Business Area	Potential Impact of the Current Gap
Supplier Performance	Underperforming or increasingly risky suppliers may not be identified in time to support renegotiation or corrective action.
Delivery Performance	Delivery delays are recognized only after they occur, limiting the ability to intervene proactively.
Inventory Decisions	Warehouses or products carrying inventory risk may not be identified until the issue is already visible operationally.
Operational Efficiency	Manual, ad hoc analysis consumes staff time that could otherwise be spent acting on findings.
Cost Visibility	Cost drivers across products, suppliers, and transportation modes remain unclear, limiting targeted cost control.
Profitability	Profit concentration and erosion are not visible by product, supplier, or region, limiting prioritization of the most profitable areas.
Management Decision-Making	Decisions rely on fragmented or manually compiled information rather than a consistent, shared evidence base.
Supply-Chain Risk	Emerging risk patterns (e.g., recurring delays tied to a specific supplier, warehouse, or region) may go unnoticed until they have recurred multiple times.

6. Business Need

The organization needs an integrated analytical decision-support capability that can:

- Monitor supply-chain performance across suppliers, products, warehouses, and transportation modes.
- Evaluate suppliers on a consistent, comparable basis.
- Identify fulfillment problems, including delayed and cancelled orders.
- Analyze delivery and transportation performance.
- Monitor inventory levels and flag potential inventory risk.
- Analyze cost and profitability across the business.
- Track KPIs using standardized, documented definitions.
- Identify emerging operational risks.
- Support predictive, forward-looking decision-making.
- Support management action through evidence-based recommendations.

7. Project Purpose

The purpose of this project is to convert the organization's existing historical supply-chain data into an integrated business decision-support capability — combining Business Analysis, structured data analysis (Excel, Power Query, SQL, Python), management reporting (Power BI), and AI/ML — so that supply-chain and procurement stakeholders can monitor performance, diagnose issues, and act proactively rather than reactively.

8. Business Objectives

- Improve supply-chain visibility — deliver a single reporting view covering all suppliers, warehouses, and transportation modes represented in the dataset.
- Improve supplier performance monitoring — enable comparison of all 10 suppliers on delivery, delay, and cancellation indicators.
- Identify delivery and fulfillment issues — provide delay and cancellation breakdowns across supplier, warehouse, transportation mode, and region.
- Improve inventory decision-making — apply a documented inventory-risk indicator across warehouse/product combinations present in the data.
- Improve cost and profitability visibility — provide cost and profit breakdowns by product, category, supplier, and region.
- Establish standardized KPIs — document and apply a consistent KPI catalog used across all reporting.
- Support proactive risk management — provide a delivery-risk indicator evaluated with Precision, Recall, F1, and ROC-AUC.
- Support demand planning — provide a demand forecast by product category with a documented error metric.

9. Stakeholder Overview

Stakeholder roles are drawn from the project's Stakeholder Register and Power–Interest Matrix. Full Power/Influence and Interest classification is documented separately in that deliverable.

Role	Interest	Business Need	Expected Value
Executive Sponsor	Overall project success, funding, and alignment with organizational priorities.	Confidence that the investment delivers measurable visibility into supply-chain performance.	Ability to prioritize and fund improvement actions based on evidence rather than anecdote.
Supply Chain Manager	Consolidated visibility into supplier, logistics, and fulfillment performance.	A reliable way to compare performance across suppliers, warehouses, and transportation modes.	Faster identification of underperforming areas and a stronger basis for operational decisions.
Procurement Manager	Supplier performance and sourcing-relevant insight.	Evidence to support supplier review and sourcing discussions.	Objective, repeatable supplier comparisons rather than informal assessment.
Inventory/Warehouse Manager	Inventory-risk visibility at the warehouse and product level.	An indicator that flags potential inventory issues before they escalate.	Earlier, more targeted intervention on inventory risk.
Logistics Manager	Delivery and transportation-mode performance.	Visibility into where and why delivery delays occur.	Evidence-based decisions on transportation-mode selection and delay mitigation.
Finance Manager	Cost and profitability visibility.	A consistent, traceable view of cost drivers and profit concentration.	More confident financial input into supply-chain decisions.
Operations Manager	End-to-end operational performance across fulfillment, delivery, and cost.	A unified operational view spanning multiple functions.	Ability to prioritize cross-functional improvement actions.
Business Analyst	Requirements traceability and successful translation of business need into the delivered solution.	Clear, validated requirements and stakeholder alignment throughout the project.	A solution that demonstrably meets the business need it was built for.
Data Analyst	Accurate, well-defined analytical output.	Clear KPI definitions and validated data to work from.	Confidence that analysis and KPI calculations are correct and defensible.
Data Scientist / ML Analyst	Well-scoped, well-understood predictive use cases.	Clear business definitions of delivery risk and demand, and honest evaluation expectations.	Models that are used and trusted because their purpose and limitations are clearly communicated.
BI Developer / Power BI Analyst	Clear KPI definitions and stakeholder reporting needs.	Validated requirements for what the report must show and to whom.	A report that stakeholders actually use because it answers their questions.

Role	Interest	Business Need	Expected Value
IT / Data Engineering Team	Manageable, well-scoped data-access requirements.	Clarity on what data access and environment support the project requires.	Reduced ad hoc requests through a clearly defined, one-time data-access scope.
Business Users / Operations Teams	Usable insight relevant to daily operational work.	Reporting that is understandable without specialized analytical training.	Practical, day-to-day decision support rather than a purely technical tool.

10. Current-State Challenges

- Fragmented visibility — supplier, delivery, inventory, and cost information exist as raw transaction data rather than a consolidated view.
- Difficulty comparing supplier performance consistently across delivery, delay, and cancellation outcomes.
- Limited visibility of delay patterns — delays are not analyzed by supplier, warehouse, transportation mode, or region in a structured way.
- Lack of centralized KPI reporting — no standardized, documented KPI set currently exists.
- Difficulty identifying inventory-risk patterns — no defined indicator currently exists for this purpose.
- Limited predictive capability — no forward-looking indicators for delivery risk or demand currently exist.
- Manual analytical effort — analysis, where it happens, is ad hoc and not easily repeated period over period.
- Difficulty converting historical data into actionable insight — data is captured but not translated into recommendations for action.

11. Future-State Vision

In the future state, business users move from raw operational data to a recommended action through a consistent, repeatable path:

Operational Data	Integrated Analysis	KPI Monitoring	Performance Insight	Risk Identification	Predictive Insight	Management Decision	Business Action
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From a business-user perspective, this means: supply-chain and procurement stakeholders will be able to open a single report, see current performance against agreed KPIs, understand where risk is emerging, review a documented predictive signal for delivery risk and demand, and act on a specific, evidence-based recommendation — rather than assembling this picture manually from raw data.

12. Business Requirements

Each requirement below is traceable to the Business Objectives in Section 8. Priority reflects the MoSCoW classification detailed in Section 13.

BR ID	Business Requirement	Business Rationale	Stakeholder	Priority	Business Outcome
BR-001	Provide a centralized, consolidated view of supply-chain performance across suppliers, products, warehouses, transportation modes, and customer regions.	No centralized view currently exists; stakeholders must work from raw, disconnected transaction data.	Supply Chain Manager; Executive Sponsor	Must Have	Stakeholders can view overall supply-chain performance from a single, consistent source.
BR-002	Provide supplier-level performance indicators covering order volume, delivery outcomes, delay frequency, and cancellation frequency.	Enables consistent, evidence-based comparison of suppliers rather than informal assessment.	Procurement Manager; Supply Chain Manager	Must Have	Suppliers can be ranked and compared on defined performance indicators.
BR-003	Provide supplier-level cost and profitability indicators derived from order-level cost and profit data.	Supplier evaluation should weigh financial impact alongside delivery performance.	Procurement Manager; Finance Manager	Should Have	Supplier decisions can consider cost/profit contribution, not delivery performance alone.
BR-004	Track order fulfillment status (Delivered, In Transit, Delayed, Cancelled) and status trends over time.	Fulfillment reliability is not currently visible in a structured, repeatable way.	Operations Manager	Must Have	Management can monitor fulfillment reliability on an ongoing basis.
BR-005	Provide fulfillment status breakdowns by supplier, product category, and customer region.	Identifies where fulfillment issues concentrate, supporting targeted action.	Operations Manager; Supply Chain Manager	Must Have	Fulfillment issues can be traced to specific suppliers, products, or regions.
BR-006	Report delivery time by supplier, warehouse, transportation mode, and customer region.	Delivery delays currently cannot be traced to a specific operational cause.	Logistics Manager; Operations Manager	Must Have	Delivery-performance issues can be diagnosed by dimension.
BR-007	Compare delivery performance and cost across transportation modes (Road, Air, Sea, Rail).	Supports mode-selection decisions that balance delivery speed against cost.	Logistics Manager	Should Have	Transportation-mode decisions can be evidence-based rather than habitual.
BR-008	Identify patterns associated with delayed and cancelled orders across supplier, warehouse, transportation mode, and region.	Enables proactive risk management rather than a purely reactive response to disruptions.	Operations Manager; Logistics Manager	Must Have	Recurring delay/cancellation patterns can be identified and addressed.

BR ID	Business Requirement	Business Rationale	Stakeholder	Priority	Business Outcome
BR-009	Provide visibility into inventory levels by warehouse and product category.	Inventory positions are not currently visible in a structured, comparable way.	Inventory/Warehouse Manager	Must Have	Inventory levels can be reviewed consistently across warehouses and products.
BR-010	Define and apply a documented inventory-risk indicator to flag warehouse/product combinations of concern.	Supports earlier, more targeted intervention on potential inventory issues.	Inventory/Warehouse Manager	Should Have	Warehouses/products carrying elevated inventory risk can be identified proactively.
BR-011	Break down total cost and shipping cost by product, category, supplier, and transportation mode.	Cost drivers are not currently visible at a level that supports targeted cost control.	Finance Manager; Supply Chain Manager	Must Have	Major cost drivers can be identified and prioritized for action.
BR-012	Report profit and profit concentration by product, category, supplier, and customer region.	Profitability patterns are not currently visible in a structured way.	Finance Manager; Executive Sponsor	Must Have	Profit contribution and risk areas can be identified by dimension.
BR-013	Report cost and profitability trends across the historical period covered by the dataset.	Supports identification of period-over-period patterns relevant to planning.	Finance Manager; Executive Sponsor	Could Have	Financial trend context is available to support planning discussions.
BR-014	Define and document a standardized set of supply-chain KPIs, each with a documented definition and calculation method.	Without standardized definitions, stakeholders may interpret performance inconsistently.	Business Analyst; All Operational Managers	Must Have	All stakeholders reference the same KPI definitions and calculations.
BR-015	Provide a management-level report highlighting trends, exceptions, risks, and performance issues across supply-chain KPIs.	Supports recurring management review with a single, consistent source of truth.	Executive Sponsor; All Managers	Must Have	Management can review supply-chain performance without manual data assembly.
BR-016	Enable drill-down within management reporting by supplier, product, warehouse, region, and time period.	Supports root-cause exploration beyond high-level summary figures.	Business Analyst; Operations Manager	Should Have	Stakeholders can investigate the detail behind a reported trend or exception.
BR-017	Provide a predictive indicator of the likelihood that an order will be delayed, based on historical order attributes.	Enables proactive intervention before a delay occurs rather than after.	Logistics Manager; Operations Manager; Data Scientist/ML Analyst	Should Have	Orders at elevated delay risk can be flagged for proactive attention.

BR ID	Business Requirement	Business Rationale	Stakeholder	Priority	Business Outcome
BR-018	Provide a documented approach to identifying potential inventory-risk situations using historical inventory and order indicators.	Extends inventory-risk visibility beyond static, point-in-time levels.	Inventory/Warehouse Manager; Data Scientist/ML Analyst	Should Have	Inventory-risk situations can be flagged using a consistent, documented method.
BR-019	Forecast future order volume by product category and time period using historical order quantities.	Supports procurement and inventory planning ahead of anticipated demand.	Supply Chain Manager; Procurement Manager	Should Have	Planning decisions can be informed by a documented demand estimate.
BR-020	Translate analytical findings into a documented set of business recommendations tied to specific evidence.	Ensures the project produces decisions and action, not only reports.	Executive Sponsor; Business Analyst	Must Have	Management receives specific, evidence-based recommendations rather than raw findings alone.
BR-021	Validate data quality (e.g., completeness, key consistency, date-format consistency) prior to analysis and document findings.	Ensures analytical outputs are based on verified, reliable data.	Business Analyst; Data Analyst	Must Have	Stakeholders can trust that reported figures are based on validated data.
BR-022	Provide a Power BI-based decision-support report as the primary interface for stakeholders to access KPIs and findings.	Consolidates deliverables into one interface stakeholders can access without technical assistance.	Executive Sponsor; All Managers	Must Have	Stakeholders have self-service access to current KPI values without requesting ad hoc analysis.
BR-023	Provide supplier risk indicators derived from historical delay and cancellation patterns to support early identification of at-risk supplier relationships.	Extends supplier performance monitoring to flag suppliers trending toward risk before they materially affect operations.	Procurement Manager; Supply Chain Manager	Should Have	Procurement can prioritize supplier reviews based on documented risk indicators.
BR-024	Provide automated highlighting of exceptions (e.g., orders exceeding typical delivery time, unusually high shipping cost) within the management report.	Adds proactive attention-directing capability beyond static reporting, subject to design and data feasibility.	Operations Manager; Business Analyst	Could Have	Exceptions can be surfaced without manual review of the full dataset.

Priority definitions: **Must Have** — required for the solution to meet its core purpose. **Should Have** — important but not release-blocking. **Could Have** — beneficial if time and data permit. See Section 13 for the complete MoSCoW prioritization, including excluded (Won't Have) capabilities.

13. MoSCoW Prioritization

Priorities were assigned based on business value, alignment with core project objectives, data availability, and implementation feasibility — not by default. The table below repeats each requirement from Section 12 with its MoSCoW priority and the specific reasoning behind that classification, and adds the capabilities explicitly excluded from this release (Won't Have), cross-referenced to Section 15 — Out of Scope.

BR ID	Business Requirement	MoSCoW Priority	Reason
BR-001	Provide a centralized, consolidated view of supply-chain performance across suppliers, products, warehouses, transportation modes, and customer regions.	Must Have	Foundational — every other requirement depends on a consolidated view existing first.
BR-002	Provide supplier-level performance indicators covering order volume, delivery outcomes, delay frequency, and cancellation frequency.	Must Have	Core to the project's stated purpose of supplier performance monitoring; fully supported by dataset fields.
BR-003	Provide supplier-level cost and profitability indicators derived from order-level cost and profit data.	Should Have	Adds financial depth to supplier evaluation; valuable but not required for a first, delivery-focused view of supplier performance.
BR-004	Track order fulfillment status (Delivered, In Transit, Delayed, Cancelled) and status trends over time.	Must Have	Order status is a core dataset field and a primary business question; essential to the minimum viable solution.
BR-005	Provide fulfillment status breakdowns by supplier, product category, and customer region.	Must Have	Directly extends BR-004 into actionable, dimension-level detail needed for targeted follow-up.
BR-006	Report delivery time by supplier, warehouse, transportation mode, and customer region.	Must Have	Delivery time is a core dataset field; central to the project's delivery-performance objective.
BR-007	Compare delivery performance and cost across transportation modes (Road, Air, Sea, Rail).	Should Have	Valuable comparative view, but secondary to establishing delivery-time visibility itself.
BR-008	Identify patterns associated with delayed and cancelled orders across supplier, warehouse, transportation mode, and region.	Must Have	Directly addresses the project's risk-identification objective using fields already available.
BR-009	Provide visibility into inventory levels by warehouse and product category.	Must Have	Inventory Level is a core dataset field; required for any inventory-related business question.
BR-010	Define and apply a documented inventory-risk indicator to flag warehouse/product combinations of concern.	Should Have	Requires a documented, agreed-upon risk definition (see Section 17, Business Rules) before it can be delivered reliably — appropriately sequenced after core visibility.

BR ID	Business Requirement	MoSCoW Priority	Reason
BR-011	Break down total cost and shipping cost by product, category, supplier, and transportation mode.	Must Have	Cost fields are core to the dataset and to the project's cost-analysis objective.
BR-012	Report profit and profit concentration by product, category, supplier, and customer region.	Must Have	Profit is a core dataset field; central to the profitability objective and to Executive Sponsor decision-making.
BR-013	Report cost and profitability trends across the historical period covered by the dataset.	Could Have	Adds useful context but is not required to answer the primary business questions; feasible only after core KPIs are stable.
BR-014	Define and document a standardized set of supply-chain KPIs, each with a documented definition and calculation method.	Must Have	Without standardized KPI definitions, every other reporting requirement risks inconsistent interpretation.
BR-015	Provide a management-level report highlighting trends, exceptions, risks, and performance issues across supply-chain KPIs.	Must Have	The primary vehicle through which stakeholders will consume all other requirements.
BR-016	Enable drill-down within management reporting by supplier, product, warehouse, region, and time period.	Should Have	Enhances usability of the core report but is not required for the report to deliver initial business value.
BR-017	Provide a predictive indicator of the likelihood that an order will be delayed, based on historical order attributes.	Should Have	High business value, but depends on a working KPI/data foundation and carries model-evaluation complexity (see Section 19, Constraints) — appropriately sequenced after Must Have items.
BR-018	Provide a documented approach to identifying potential inventory-risk situations using historical inventory and order indicators.	Should Have	Similar reasoning to BR-017: valuable but dependent on the core inventory-visibility capability (BR-009) being established first.
BR-019	Forecast future order volume by product category and time period using historical order quantities.	Should Have	Forecasting adds planning value but is not required to address the project's core visibility and risk-identification objectives.
BR-020	Translate analytical findings into a documented set of business recommendations tied to specific evidence.	Must Have	Recommendations are the mechanism that converts analysis into business action; without them, the project delivers reports but not decisions.

BR ID	Business Requirement	MoSCoW Priority	Reason
BR-021	Validate data quality (e.g., completeness, key consistency, date-format consistency) prior to analysis and document findings.	Must Have	Data-quality issues (e.g., mixed date formats) are known constraints (Section 19); validating them is a precondition for trustworthy analysis.
BR-022	Provide a Power BI-based decision-support report as the primary interface for stakeholders to access KPIs and findings.	Must Have	Power BI is the agreed reporting mechanism (technology stack) for delivering Must Have visibility and KPI requirements to stakeholders.
BR-023	Provide supplier risk indicators derived from historical delay and cancellation patterns to support early identification of at-risk supplier relationships.	Should Have	Extends supplier monitoring meaningfully but depends on core supplier performance visibility (BR-002) being in place first.
BR-024	Provide automated highlighting of exceptions (e.g., orders exceeding typical delivery time, unusually high shipping cost) within the management report.	Could Have	Desirable usability enhancement; feasibility depends on report design decisions made later in the project.
BR-025	Real-time IoT monitoring of shipments and warehouse conditions.	Won't Have	No real-time or IoT data exists in the current dataset; would require new data sources and infrastructure outside this project's scope (see Section 15).
BR-026	Direct integration with a live ERP system.	Won't Have	No ERP connectivity exists; this project analyzes a historical data extract, not a live transactional system.
BR-027	Automated procurement workflow.	Won't Have	No procurement workflow or approval data exists in the dataset; this project provides insight, not process automation.
BR-028	Supplier self-service portal.	Won't Have	Outside the analytical scope of this project; would require a separate application build.
BR-029	Automated purchase-order approval.	Won't Have	No purchase-order data exists in the dataset; approval automation is a separate operational system capability.
BR-030	Real-time inventory system integration.	Won't Have	Inventory Level in the dataset is a static snapshot, not a live feed; real-time integration is a future-state infrastructure capability.

14. Business Scope

The following business capabilities are included in this project:

- Supplier performance analysis
- Order fulfillment analysis
- Delivery and logistics (transportation-mode) analysis
- Inventory visibility and inventory-risk identification
- Cost analysis
- Profitability analysis
- Standardized KPI reporting
- Management-level reporting
- Predictive insight (delivery risk and demand forecasting)
- Business recommendations derived from analytical findings

15. Out of Scope

The following capabilities are not supported by the current project or dataset. They may represent valid future enhancements but require data or systems this project does not have access to:

- Real-time monitoring (including IoT-based tracking) of shipments, warehouses, or inventory.
- Direct integration with a live ERP system.
- Procurement workflow automation.
- Supplier contract management.
- Payment processing.
- Automated purchase-order approval.
- Real-time warehouse or inventory systems integration.
- Any capability dependent on fields not present in the dataset, including Purchase Order Number, Procurement Approval Date, Supplier Contract Value, Supplier Payment Terms, Negotiated Price, Reorder Point, Safety Stock, or Supplier SLA.

These items are documented as Future-State Requirements. They are excluded here because the necessary data or system access does not currently exist — not because they lack business value.

16. High-Level Business Process

The business-level flow supported by this solution is as follows:

Operational Data	Data Preparation	Supply Chain Analysis	KPI Monitoring	Issue/Risk Identification	Predictive Analysis	Management Insight	Decision	Recommended Action
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This is a business-process flow, not a technical architecture. Technical design (data pipelines, model architecture, report structure) is addressed in later, separate project phases and, where applicable, the Functional Requirements Document (FRD).

17. High-Level Business Rules

Rule ID	Business Rule
BRU-01	KPI definitions must be documented before being used in reporting.
BRU-02	Order statuses must follow the approved values: Delivered, In Transit, Delayed, Cancelled.
BRU-03	Invalid or incomplete data must be identified and documented during data preparation.
BRU-04	Derived metrics (e.g., an inventory-risk indicator) must have a documented, agreed-upon formula or definition.
BRU-05	Predictive outputs must clearly identify themselves as predictions, not confirmed outcomes.
BRU-06	Business decisions should not rely solely on predictive model output; predictions are one input among others.
BRU-07	Any inventory-risk definition must be documented and agreed with the Inventory/Warehouse Manager before use in reporting.

18. Assumptions

ID	Assumption	Reason	Business Impact if Incorrect
AS-01	Each row represents one order transaction, not a line item within a larger order.	Order ID is unique across all 10,000 rows.	Quantity/cost aggregation by "order" could be misstated if rows actually represent order lines.
AS-02	Order Status reflects status as of the data extract date, not a full status-change history.	The dataset contains a single status value per order with no change timestamp.	Any analysis of time-to-delay or status transitions beyond the given dates would be unsupported.
AS-03	Order/Shipment/Delivery Date use either M/D/YYYY or MM-DD-YYYY format, both representing month-before-day.	Both patterns are observed in the raw file; no field indicates which applies to which row.	Date-based calculations (e.g., delivery time) could be incorrect for affected rows if the convention differs.
AS-04	Delivery Time (Days) is consistent with Delivery Date minus Shipment Date.	The field is provided directly rather than derived by this project.	Delay analysis could be mislabeled if the source calculation used a different date pair.
AS-05	Inventory Level represents a warehouse/product snapshot associated with the order, not a running real-time balance.	No timestamp or explicit definition is provided beyond the numeric value per row.	Inventory-risk indicators built on this field would need redefinition if it represents something else (e.g., capacity).

ID	Assumption	Reason	Business Impact if Incorrect
AS-06	Product ID values are transaction-specific rather than a stable identifier for a single product.	The same Product Name maps to thousands of distinct Product ID values.	Product-level analysis by Product ID rather than Product Name would fragment results incorrectly.
AS-07	Currency for all monetary fields is a single, consistent currency across all records.	No currency field or symbol is present in the dataset.	Cost/profit comparisons would be invalid without conversion if multiple currencies are actually mixed.
AS-08	Profit is a per-order net figure already net of relevant costs captured in the dataset.	Profit is provided directly, including negative values, implying costs are already applied.	Profitability KPIs would need adjustment if Profit excludes certain costs (e.g., shipping).

19. Constraints

ID	Constraint	Business Impact	Mitigation
CO-01	Historical data only — January 2024 to December 2025, no live or streaming feed.	Findings and models reflect only this historical window and may not capture current conditions.	Clearly label all outputs as historical analysis; treat real-time monitoring as a future-state requirement.
CO-02	Mixed date formats within the same columns.	Naive parsing could silently misread dates, corrupting delivery-time or trend analysis.	Apply and validate explicit date-parsing logic before any date-based analysis (BR-021).
CO-03	Product ID is not a stable product key.	Product-level analysis by Product ID would fragment results incorrectly.	Use Product Name and Product Category as the stable product dimension throughout.
CO-04	Procurement-specific fields (PO Number, Contract Value, SLA, Lead Time, etc.) are not present.	Several natural procurement KPIs cannot be calculated from this data.	Scope these out explicitly (Section 15) and label them as future-state requirements — never fabricated.
CO-05	No connection to a live ERP, WMS, or procurement system.	The project cannot demonstrate live monitoring, alerting, or system integration.	Scope the project as historical/batch analysis; note real-time integration as future-state.
CO-06	AI/ML components depend on historical patterns.	Model performance is not guaranteed to hold if business conditions change materially.	State this limitation explicitly in model documentation and management reporting (BRU-05, BRU-06).
CO-07	Order Status is imbalanced (82.7% Delivered, 8.2% In Transit, 5.9% Delayed, 3.2% Cancelled).	Naive accuracy-based evaluation of the delivery-risk model could be misleading.	Use Precision, Recall, F1, and ROC-AUC rather than accuracy alone to evaluate delivery-risk prediction.

ID	Constraint	Business Impact	Mitigation
CO-08	No separate supplier, product, warehouse, or customer master tables exist.	Some dimension attributes must be reconstructed from the flat transaction file.	Document the dimension-extraction logic used during data preparation.
CO-09	No currency indicator is present on any monetary field.	Cross-region cost/profit comparisons assume a single, consistent currency (AS-07).	State this assumption explicitly wherever monetary comparisons are shown.
CO-10	Analysis is limited to this single dataset; no external data sources are used.	Root-cause findings are limited to relationships observable within the available fields.	Frame root-cause findings as associations within the available data, not externally validated causes.

20. Dependencies

- Continued availability and integrity of the source dataset (SupplyChain_Analytics_10000.csv).
- Data quality sufficient to support the analysis described in this document (see Section 19, Constraints).
- Timely stakeholder validation of requirements, KPI definitions, and analytical findings.
- Agreement on approved KPI definitions before reporting is finalized (BRU-01).
- Sufficient historical records to support meaningful pattern analysis and predictive modeling.
- Availability of the agreed analytical tools (Excel/Power Query, SQL, Python, Power BI) throughout the project.
- Business validation of predictive outputs (delivery risk, demand forecast) before they inform decisions.

21. Success Criteria

Success is defined in observable, business terms rather than as invented improvement percentages:

- Stakeholders can monitor agreed supply-chain KPIs from a centralized reporting solution.
- Supplier performance can be compared consistently using defined indicators.
- Delayed and cancelled orders can be identified and traced to relevant suppliers, warehouses, or transportation modes.
- Inventory-risk situations can be highlighted using an approved, documented indicator.
- Cost and profitability drivers can be identified by product, category, supplier, and region.
- Predictive models (delivery risk, demand forecast) provide documented evaluation metrics.
- Business recommendations are supported by analytical evidence and traceable to specific findings.

22. Expected Business Benefits

- Improved supply-chain visibility across suppliers, products, warehouses, and transportation modes.
- Better, more consistent supplier management.
- Earlier risk identification, including delivery delays and cancellations.

- Improved delivery monitoring.
- Improved inventory decisions through a defined inventory-risk indicator.
- Improved cost visibility.
- Stronger, more consistent management reporting.
- Data-driven decision-making, supported by a recurring reporting framework.
- More proactive supply-chain management, supported by predictive insight.

23. High-Level Risks

Risk ID	Risk	Impact	Mitigation
RSK-01	Poor data quality (e.g., unresolved date-format inconsistencies).	Analytical outputs could be inaccurate or misleading.	Apply and document a data-quality validation step before analysis (BR-021).
RSK-02	Insufficient historical data for reliable prediction in minority classes.	Predictive models may have limited accuracy for less-represented outcomes (e.g., Delayed, Cancelled orders).	Report evaluation metrics transparently; use Precision/Recall/F1/ROC-AUC rather than accuracy alone.
RSK-03	Misleading predictive results.	Stakeholders could over-rely on predictions without understanding their limitations.	Clearly label predictive outputs as estimates (BRU-05); require decisions not to rely solely on model output (BRU-06).
RSK-04	Inconsistent KPI definitions.	Stakeholders may interpret the same metric differently, undermining trust in reporting.	Document and agree KPI definitions before reporting is finalized (BR-014, BRU-01).
RSK-05	Stakeholder adoption challenges.	Reporting and recommendations may not be used in actual decision-making.	Involve key stakeholders in requirements validation and UAT before rollout.
RSK-06	Scope expansion.	Requests for capabilities dependent on unavailable data could pull the project beyond its defined scope.	Reference the Out of Scope register (Section 15) when evaluating new requests.

24. Requirement Priority Summary

Priority	Count
Must Have	14
Should Have	8
Could Have	2
Won't Have (this release)	6
Total Requirements Considered	30

These counts match the Business Requirements table (Section 12: BR-001–BR-024) and the Won't Have items documented in Section 13 (BR-025–BR-030).

25. Approval / Sign-off

This Business Requirements Document requires review and approval from the following stakeholder roles prior to proceeding to the next project phase.

Stakeholder Role	Approval Status	Approval Date	Comments
Executive Sponsor	Pending		
Supply Chain Manager	Pending		
Procurement Manager	Pending		
Operations Manager	Pending		
Finance Manager	Pending		
Business Analyst (Prepared By)	Pending		

Approval Date and Comments fields are intentionally left blank for completion by the relevant stakeholder. Role names are used throughout this document; no individual names are fabricated.

— End of Business Requirements Document —

This BRD defines what the business needs and why. A separate Functional Requirements Document (FRD) will later define how the solution must function.